A Universal Square-Root-of-Six Bound for Kissing Numbers

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Abstract

Let τ_n denote the kissing number in Euclidean dimension n. We prove that

$$\tau_n \le (\sqrt{6})^n \qquad (n \ge 1).$$

The proof packs explicit bodies of revolution inside the unit ball. An inscribed bicone gives the estimate in dimension two and every dimension at least five; slightly larger bodies suffice in dimensions three and four. The body volumes are obtained by one-dimensional integration. Since $\tau_2 = 6$, the base $\sqrt{6}$ is optimal for a bound of this form valid in every dimension. The argument, including its geometric, analytic, and numerical steps, is formalized in Lean 4/Mathlib.

AI provenance and verification status. The mathematical argument, manuscript, and Lean 4/Mathlib formalization were generated with OpenAI's GPT-6 Astra, from prompts supplied by the author. The formalization was registered in Palomar on September 5, 2026, as `PALOMAR-2026-09-05-000002`, version 1, following successful mechanical verification and an automated review that identified no problems. Registration is not human peer review or a novelty certificate. The manuscript has not undergone independent human mathematical review.

1 Statement of the bound

A *kissing configuration* in $\mathbb{R}^n$ is a finite set X satisfying

$$\|x\| = 1 \quad (x \in X), \qquad \langle x, z\rangle \le \frac{1}{2} \quad (x, z \in X,\ x \ne z). \tag{1}$$

Multiplying its vectors by two gives the centers of nonoverlapping unit balls touching the unit ball at the origin. We write τ_n for the largest cardinality of a kissing configuration in $\mathbb{R}^n$. The finite bound proved below also establishes the existence of this maximum.

Theorem 1. *For every integer $n \ge 1$,*

$$\tau_n \le (\sqrt{6})^n.$$

Corollary 2. *The least real number $\alpha > 0$ such that*

$$\tau_n \le \alpha^n \qquad \text{for every } n \ge 1$$

is $\sqrt{6}$. Equivalently,

$$\sup_{n\ge 1} \tau_n^{1/n} = \sqrt{6}.$$

The point is the optimal base in a bound that holds in every dimension with no additional prefactor. No claim of asymptotic optimality is intended.

2 Proof

Throughout, put $s = \sqrt{3}$ and $a = s/2$. Let B_n be the open unit ball in $\mathbb{R}^n$, and write $V_n = \mathrm{vol}(B_n)$ for its Lebesgue volume.

2.1 Disjoint cones and a volume comparison

For a unit vector x, define the open cone

$$C_x = \{y \in \mathbb{R}^n : \langle x, y\rangle > a\|y\|\}.$$

Thus C_x has axis x and half-angle $\pi/6$.

Lemma 3. *Let X be a kissing configuration in $\mathbb{R}^n$, where $n \geq 2$, and let e be a fixed unit vector. If K is a measurable subset of $C_e \cap B_n$ with volume q, then*

$$|X|q \leq V_n. \tag{2}$$

Proof. We first check that $C_x \cap C_z = \varnothing$ for distinct $x, z \in X$. Otherwise, normalizing a vector in the intersection gives a unit vector u with

$$b = \langle x, u\rangle > a, \qquad c = \langle z, u\rangle > a.$$

Write $x = bu + p$ and $z = cu + r$, where $p, r \perp u$. Then

$$\|p\|^2 = 1 - b^2 < \frac{1}{4}, \qquad \|r\|^2 = 1 - c^2 < \frac{1}{4}.$$

Cauchy–Schwarz now gives

$$\langle x, z\rangle = bc + \langle p, r\rangle \geq bc - \|p\|\|r\| > \frac{3}{4} - \frac{1}{4} = \frac{1}{2},$$

contrary to (1).

For each $x \in X$, choose an orthogonal transformation sending e to x. Its image K_x of K lies in $C_x \cap B_n$ and has volume q. These measurable images are pairwise disjoint, so finite additivity and inclusion in B_n give (2). $\square$

We construct the bodies in orthogonal coordinates $(t, v) \in \mathbb{R} \times \mathbb{R}^{n-1}$, with $e = (1, 0)$. In these coordinates,

$$(t, v) \in C_e \iff t > 0 \text{ and } 3\|v\|^2 < t^2, \qquad (t, v) \in B_n \iff t^2 + \|v\|^2 < 1. \tag{3}$$

For a continuous nonnegative function g on $[b, c]$, Fubini's theorem and scaling of Lebesgue measure give

$$\mathrm{vol}\{(t, v) : b < t < c,\ \|v\| < g(t)\} = V_{n-1} \int_b^c g(t)^{n-1}\, dt. \tag{4}$$

Indeed, the section at t is an $(n-1)$-dimensional ball of radius $g(t)$. Every body used below is a finite union of such measurable sets, with disjoint intervals for the first coordinate.

2.2 The bicone bound

For $n \geq 2$, consider

$$L_n = \{(t, v) : 0 < t < a,\ \|v\| < t/s\},$$
$$U_n = \{(t, v) : a < t < 1,\ \|v\| < (2 + s)(1 - t)\}.$$

Their union $K_n = L_n \cup U_n$ is a bicone, with the common section at $t = a$ omitted. Both profiles have radius $1/2$ at that section.

Lemma 4. *The body K_n lies in $C_e \cap B_n$ and has volume*

$$\mathrm{vol}(K_n) = \frac{V_{n-1}}{n2^{n-1}}. \tag{5}$$

Consequently every kissing configuration $X \subset \mathbb{R}^n$ satisfies

$$|X| \le n2^{n-1}\frac{V_n}{V_{n-1}}. \tag{6}$$

Proof. For $(t, v) \in L_n$, one has $3\|v\|^2 < t^2$ and

$$t^2 + \|v\|^2 < \frac{4}{3}t^2 < \frac{4}{3}a^2 = 1.$$

For the upper profile $h(t) = (2+s)(1-t)$, the identities

$$sh(a) = a, \qquad (2+s)^2(1-a) = 1+a$$

show, for $a < t < 1$, that

$$sh(t) < t, \qquad h(t)^2 < 1 - t^2.$$

Here the first inequality compares a decreasing affine function with t, and the second follows by multiplying $(2+s)^2(1-t) < 1+t$ by $1-t > 0$. Thus (3) puts both pieces inside $C_e \cap B_n$.

The two integrals in (4) are

$$\int_0^a (t/s)^{n-1}\,dt = \frac{s}{n2^n},$$
$$\int_a^1 \big((2+s)(1-t)\big)^{n-1}\,dt = \frac{2-s}{n2^n}.$$

For the second equality, use $(2+s)(1-a) = 1/2$. Adding the integrals proves (5); Lemma 3 then proves (6). $\square$

We use the standard unit-ball volume formula and its recurrence:

$$V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}, \qquad V_{d+2} = \frac{2\pi}{d+2}V_d \quad (d \ge 1). \tag{7}$$

The recurrence follows directly from $\Gamma(z+1) = z\Gamma(z)$. In particular,

$$\begin{gathered} V_1 = 2, \qquad V_2 = \pi, \qquad V_3 = \frac{4\pi}{3}, \qquad V_4 = \frac{\pi^2}{2}, \\ V_5 = \frac{8\pi^2}{15}, \qquad V_6 = \frac{\pi^3}{6}, \qquad V_7 = \frac{16\pi^3}{105}. \end{gathered} \tag{8}$$

The estimates for π needed here are implied by $0 < \pi < 22/7$.

Lemma 5. *For every integer $n \ge 6$,*

$$V_n \le V_{n-1}, \qquad n2^{n-1} \le (\sqrt{6})^n.$$

Proof. The first inequality holds at $n = 6$, since $V_6/V_5 = 5\pi/16 < 1$, and at $n = 7$, since $V_7/V_6 = 32/35 < 1$. If it holds two dimensions earlier, then (7) gives

$$V_n = \frac{2\pi}{n}V_{n-2} \le \frac{2\pi}{n}V_{n-3} \le \frac{2\pi}{n-1}V_{n-3} = V_{n-1}.$$

This proves the first assertion by induction separately on even and odd dimensions.

For the second assertion, the initial cases are

$$6 \cdot 2^5 = 192 < 216 = (\sqrt{6})^6, \qquad 7 \cdot 2^6 = 448 < 216\sqrt{6} = (\sqrt{6})^7;$$

the latter inequality follows by squaring $56/27 < \sqrt{6}$. If the assertion holds at $n-2$ and $n \geq 8$, then $4n \leq 6(n-2)$ gives

$$n2^{n-1} = 4n2^{n-3} \leq 6(n-2)2^{n-3} \leq 6(\sqrt{6})^{n-2} = (\sqrt{6})^n.$$

□

Together, Lemmas 4 and 5 prove the required cardinality bound for every $n \geq 6$. The bicone also suffices in dimensions two and five. Indeed, (6) and (8) give

$$n = 2: \quad |X| \leq 2\pi < 7, \qquad n = 5: \quad |X| \leq \frac{256}{3} < 86. \tag{9}$$

Since $|X|$ is an integer, these bounds imply respectively $|X| \leq 6 = (\sqrt{6})^2$ and $|X| \leq 85 < 36\sqrt{6} = (\sqrt{6})^5$. The last strict inequality follows from $85^2 < 36^2 \cdot 6$.

2.3 Dimension three

In dimension three, retain L_3 and replace the upper part of the bicone by

$$W_3 = \{(t, v) : a < t < 1,\ \|v\| < \sqrt{1 - t^2}\}.$$

This body is in B_3 by construction. Moreover, $t > a$ implies $3(1 - t^2) < t^2$, so (3) places W_3 in C_e as well. By (4),

$$\begin{aligned}
\mathrm{vol}(L_3 \cup W_3) &= V_2 \left(\frac{s}{24} + \int_a^1 (1 - t^2)\, dt \right) \\
&= \pi \left(\frac{s}{24} + \frac{2}{3} - \frac{3s}{8} \right) = \frac{(2 - s)\pi}{3}.
\end{aligned}$$

Since $s < 26/15$, as follows by squaring, this volume is strictly larger than $4\pi/45 = V_3/15$. Lemma 3 therefore yields $|X| < 15$, hence

$$|X| \leq 14 < 6\sqrt{6} = (\sqrt{6})^3. \tag{10}$$

Here the final comparison follows from $14^2 < 36 \cdot 6$.

2.4 Dimension four

For dimension four, a polynomial upper profile makes the volume calculation elementary. Set

$$p(t) = (1 - t)(14t + 2 - 6s), \qquad W_4 = \{(t, v) : a < t < 1,\ \|v\| < p(t)\}.$$

We shall prove that $L_4 \cup W_4$ lies in $C_e \cap B_4$ and has volume greater than $V_4/37$.

First, $p(a) = 1/2$, $p(1) = 0$, and $p(t) \geq 0$ on $[a, 1]$, since $14t + 2 - 6s \geq s + 2 > 0$. To check containment in the ball, define

$$Q(t) = 196t^2 - (140 + 70s)t - 48 + 74s.$$

A direct factorization using $s^2 = 3$ gives

$$1 - t^2 - p(t)^2 = (1 - t)(t - a)Q(t). \tag{11}$$

For $t \geq a$, the identity

$$Q(t) = 4s - 6 + (t-a)\big(196(t+a) - 140 - 70s\big)$$

shows that $Q(t) > 0$: indeed $s > 3/2$, and the expression in parentheses is at least $126s - 140 > 0$. Thus

$$p(t)^2 \leq 1 - t^2 \qquad (a \leq t \leq 1).$$

Every point of W_4 therefore lies in B_4. As in dimension three, $t > a$ then implies $3\|v\|^2 < t^2$, so $W_4 \subset C_e$.

It remains to compute its volume. Expanding $p(t)^3$ and integrating its monomials gives

$$\int_a^1 p(t)^3\,dt = \frac{33169 - 19150s}{40}. \tag{12}$$

For an explicit evaluation, put $\delta = 1 - a$ and $A = 16 - 6s$. The substitution $u = 1 - t$ gives $p(t) = u(A - 14u)$, and the left side of (12) equals

$$\frac{A^3\delta^4}{4} - \frac{42A^2\delta^5}{5} + 98A\delta^6 - 392\delta^7.$$

Substituting $\delta = 1 - s/2$ and reducing powers with $s^2 = 3$ yields (12). Adding the lower part, whose profile integral is $s/64$, gives

$$\mathrm{vol}(L_4 \cup W_4) = TV_3, \qquad T = \frac{265352 - 153195s}{320}. \tag{13}$$

All that is needed is a rational lower estimate for T. The inequality $s < 1351/780$ follows from $1351^2 - 3 \cdot 780^2 = 1$. Consequently

$$T > \frac{541}{16640} > \frac{33}{1036} = \frac{3}{296}\,\frac{22}{7} > \frac{3\pi}{296}.$$

Using (8), we obtain

$$\mathrm{vol}(L_4 \cup W_4) > \frac{3\pi}{296}\,\frac{4\pi}{3} = \frac{\pi^2}{74} = \frac{V_4}{37}.$$

Lemma 3 now yields $|X| < 37$. Integrality gives

$$|X| \leq 36 = (\sqrt{6})^4. \tag{14}$$

2.5 Completion and optimality

Proof of Theorem 1. Let X be any kissing configuration in $\mathbb{R}^n$. For $n = 1$, every unit vector is 1 or -1, so $|X| \leq 2 < \sqrt{6}$. Dimensions two and five are covered by (9), dimensions three and four by (10) and (14), and every $n \geq 6$ by Lemmas 4 and 5. Thus $|X| \leq (\sqrt{6})^n$ in every positive dimension.

The realizable cardinalities form a nonempty subset of

$$\{0, \ldots, \lfloor(\sqrt{6})^n\rfloor\}.$$

This finite set of cardinalities has a largest element, realized by a configuration. This element is τ_n, and it satisfies the same bound. $\square$

Proof of Corollary 2. The six points

$$(1,0), \quad (\tfrac{1}{2}, \tfrac{s}{2}), \quad (-\tfrac{1}{2}, \tfrac{s}{2}), \quad (-1,0), \quad (-\tfrac{1}{2}, -\tfrac{s}{2}), \quad (\tfrac{1}{2}, -\tfrac{s}{2})$$

are distinct unit vectors, and the inner product of any two distinct members is at most $1/2$. They form a kissing configuration. Theorem 1 gives the reverse bound, so $\tau_2 = 6$.

The base $\sqrt{6}$ works by Theorem 1. Any positive universal base α must satisfy $6 = \tau_2 \leq \alpha^2$, hence $\alpha \geq \sqrt{6}$. Likewise each $\tau_n^{1/n} \leq \sqrt{6}$, with equality at $n = 2$, which proves the supremum statement. $\square$

3 Lean certification and Palomar registration

The proof in Section 2 is formalized in the accompanying repository:

`https://github.com/sebastienpalcoux/sqrt6-kissing-bound`.

The certificate checks cone separation, measurable disjoint packing, body-volume integrals, ball-volume estimates, and every dimension case. It also proves that the kissing number is an attained maximum and verifies the six-point configuration used for optimality. Standard analysis comes from proved Mathlib theorems.

The eight principal declarations selected by `comparator.json` were registered as

`PALOMAR-2026-09-05-000002`, version 1.

They express the universal bound, the existence of attained kissing numbers, the planar value six, and the equivalent optimality statements. The registered repository commit is

`cc87870ad897f0fe15e3cdffd3d7acd15f0c1ba9`.

Comparator checked the correspondence between the independently stated Challenge and its Solution, and both Lean's kernel and the independent NanoDa checker verified the exported proofs. Palomar's automated editorial review identified no problems with the statements, definitions, presentation, literature account, or research interest. These checks do not constitute independent human review or establish novelty.

The registered development pins Lean 4.33.0 and Mathlib revision

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The registration preserves that exact source snapshot. This later revision updates the disclosure and registration information; the mathematical argument and Lean source are unchanged.

After obtaining the dependencies with `lake exe cache get`, the commands

```
bash scripts/check.sh
bash scripts/verify-comparator.sh
```

respectively build the certificate and audit its axioms, and repeat the Comparator and NanoDa checks. Only Lean's standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound` are allowed in the proof development. The isolated Challenge contains eight deliberate statement placeholders, whose replacements are verified by Comparator; the Solution and substantive proofs contain no admitted results or additional axioms.

Acknowledgment

This note was prompted by the MathOverflow question "Upper bound of the kissing number in n dimensions" and by Henry Cohn's answer explaining how effective spherical-code bounds bear on the problem.

References

[1] S. Palcoux, *Upper bound of the kissing number in n dimensions*, MathOverflow question 282644 (2017), `https://mathoverflow.net/questions/282644/`.

[2] H. Cohn, *Answer to "Upper bound of the kissing number in n dimensions"*, MathOverflow answer 282685 (2017), `https://mathoverflow.net/a/282685/`.

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